

Problem-Solution Fit

Date	28 JUNE 2028
Team ID	xxxxxx
Project Name	Smart Agricultural Production Optimization Engine (OptiCrop)
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

Use the framework below to ensure your proposed solution accurately addresses the root causes, behaviors, and limitations of your target audience. Fill out the canvas in the respective boxes just like the original visual template.

1. CUSTOMER SEGMENT(S) [CS] • Farmers • Agricultural Researchers • Agricultural Officers • Policymakers • Agricultural Students	6. CUSTOMER LIMITATIONS EG. BUDGET, DEVICES [CL] • Limited technical knowledge • Lack of real-time agricultural insights • Limited internet connectivity in rural areas • Budget constraints • Lack of awareness of AI-based farming technologies	5. AVAILABLE SOLUTIONS PROS & CONS [AS] Pros • Easy to access • Expert guidance available • Low implementation cost Cons • Time-consuming • General recommendations • Not personalized to specific soil conditions • Less accurate for crop prediction.
2. PROBLEMS / PAINS + ITS FREQUENCY [PR] • Difficulty selecting the most suitable crop. • Low agricultural productivity due to poor crop planning. • Uncertainty caused by changing weather conditions. • Inefficient utilization of soil nutrients.	9. PROBLEM ROOT / CAUSE [RC] • Lack of data-driven decision making. • Insufficient knowledge about soil fertility. • Unpredictable climate conditions.	7. BEHAVIOR + ITS INTENSITY [BE] • Farmers depend on traditional farming methods. • Consult local experts before crop selection. • Search online for farming information. • Moderate adoption of digital agriculture tools.

• Frequent crop failures due to incorrect crop selection.	• Traditional crop selection methods. • Limited access to modern agricultural technologies.	• High dependence on seasonal experience.
3. TRIGGERS TO ACT [TR] • Need to maximize crop yield. • Rising farming costs. • Seasonal changes in weather. • Government initiatives promoting smart agriculture. • Demand for sustainable farming practices.	10. YOUR SOLUTION [SL] • OptiCrop is an AI-powered Smart Agricultural Production Optimization Engine that analyzes soil nutrients (Nitrogen, Phosphorus, Potassium), temperature, humidity, pH, and rainfall using machine learning algorithms. It recommends the most suitable crop for cultivation, helping farmers improve productivity, optimize resource utilization, reduce farming risks, and promote sustainable agriculture.	8. CHANNELS OF BEHAVIOR [CH] ONLINE • Agricultural websites • Government agriculture portals • Mobile farming applications • YouTube farming channels • Weather forecasting websites
4. EMOTIONS BEFORE / AFTER [EM] BEFORE: • Confused • Worried • Uncertain • Frustrated AFTER: • Confident • Satisfied • Optimistic • Motivate		OFFLINE • Agricultural extension officers • Local farming communities • Agricultural training programs • Seed and fertilizer shops